

# Controllable Text Generation via Diffusion Models on the E2E NLG Benchmark

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**Abstract.** We study controllable text generation with continuous-space diffusion language models and compare them against autoregressive baselines on the E2E NLG benchmark. Our implementation of Diffusion-LM combines four design choices that we found necessary for stable training at small scale: prefix conditioning with embedding clamping at every denoising step, target-masked losses that exclude prefix and padding positions, initialization of the token embedding table from a pretrained BERT tokenizer, and an exponential moving average (EMA) of the denoiser weights. On top of this base model we add self-conditioning, which feeds the previous  $x_0$  prediction back into the denoiser, and an attribute classifier trained jointly with the diffusion loss that provides classifier-guided sampling over the `food` slot. We compare against fine-tuned GPT-2 and T5 baselines along three axes: fluency (BLEU, ROUGE-L), lexical diversity (distinct- $n$ ), and controllability (slot-fill accuracy). Ablations isolate the contributions of noise schedule (sqrt vs. cosine), DDIM sampling step count, and prefix length. We discuss the trade-off between hard prefix clamping and soft classifier guidance, the practical limits of small-scale continuous diffusion for natural language generation, and the extent to which the heavily templated structure of E2E NLG itself favors autoregressive beam search over iterative refinement.

**Keywords:** Diffusion models · Controllable generation · Non-autoregressive generation · Classifier guidance · E2E NLG · Natural language generation.

## 1 Introduction

Autoregressive (AR) language models dominate modern text generation, but their left-to-right decoding ties attribute control to ad-hoc prompt engineering and tends to under-explore the output distribution, hurting lexical diversity. Diffusion models, which have transformed image synthesis [8,14], offer a non-autoregressive alternative that operates in a continuous embedding space and supports plug-and-play guidance [10]. The diffusion formulation iteratively refines an entire sequence in parallel, which makes attribute conditioning both more local (a single position can be clamped) and more global (gradients from an attribute classifier reach every position simultaneously).

This project implements a Diffusion-LM variant with prefix conditioning on the E2E NLG benchmark [15] and asks: *can a small, trainable-from-scratch diffusion language model match autoregressive baselines on a slot-driven generation task while improving the diversity–controllability trade-off?*

*Contributions.*

1. A from-scratch Diffusion-LM implementation with three interchangeable noise schedules (sqrt, cosine, linear), prefix-clamped denoising diffusion implicit model (DDIM) sampling [19], exponential moving average (EMA) weights, and BERT-initialized token embeddings.
2. Two controllability mechanisms in one model: *hard* prefix clamping that fixes the meaning representation throughout denoising, and a *soft* jointly trained attribute classifier used for gradient-based guidance [4] on a chosen slot.
3. Self-conditioning [1] as an architectural addition that lets the denoiser refine its own previous  $x_0$  prediction across consecutive DDIM steps.
4. A side-by-side comparison with fine-tuned GPT-2 and T5 baselines on E2E NLG using fluency (BLEU, ROUGE-L), lexical diversity (distinct- $n$ ), and controllability (slot-fill accuracy).
5. Ablations isolating the effect of noise schedule, DDIM step count, and prefix length.

## 2 Related Work

*Diffusion language models.* Diffusion-LM [10] introduced continuous-space text diffusion: tokens are mapped to learned embeddings, noise is added in embedding space according to the sqrt schedule, and a transformer denoiser predicts the clean embeddings ( $x_0$ -prediction). At inference, a rounding step projects denoised vectors back to the vocabulary. Subsequent work extended the framework to sequence-to-sequence tasks (DiffuSeq [6]) and to semi-autoregressive decoding with simplex representations (SSD-LM [7]). Our implementation follows the original Diffusion-LM closely, with the additions described in Section 3.

*Diffusion design choices.* Self-conditioning [1] feeds a stop-gradient  $x_0$  estimate from a first forward pass back into the denoiser on a second pass, letting consecutive denoising steps share information. Classifier guidance [4] steers sampling toward a target class via  $x_t \leftarrow x_t + s \nabla_{x_t} \log p_\phi(y \mid x_t)$ ; we apply it with prefix positions zeroed so the MR is not perturbed. PPLM [2] and FUDGE [20] achieve a similar effect for autoregressive decoders, but the diffusion analogue is structurally simpler — gradients reach every position at every step rather than only modifying next-token logits.

### 3 Methodology

#### 3.1 Diffusion-LM with Prefix Conditioning

Token ids  $w_{1:L}$  are embedded as  $x_0 = E[w_{1:L}] \in \mathbb{R}^{L \times d}$ , with  $d=768$ . The forward process injects Gaussian noise according to the chosen schedule:

$$q(x_t | x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) I). \quad (1)$$

We adopt the sqrt schedule of [10] as our default:  $\bar{\alpha}_t = 1 - \sqrt{t/T + s}$  with  $s=10^{-4}$  and  $T=2000$ , and ablate against cosine [14] and linear [8] schedules. A transformer denoiser  $f_\theta(x_t, t)$  with six layers, eight heads, and hidden width 512 is trained to predict  $x_0$ .

*Prefix conditioning.* The first  $P=32$  positions hold the tokenized MR (a flattened “**key: value | key: value**” string). We keep these positions at  $x_0$  in the forward process (no noise added) and exclude them from the loss. At sampling time we clamp the prefix embeddings to the MR embeddings at every DDIM step, so the denoised target is conditioned on the MR throughout. This is the diffusion analogue of inpainting in vision models, and provides controllability *by construction*: the model is never exposed to a noisy MR and cannot drift away from it.

*Target masking.* The target mask covers the entire target window ( $P < i \leq L$ ), including trailing pad tokens. Excluding pad positions from the loss — a natural early choice — creates a training–inference mismatch: training anchors pad positions to clean pad embeddings, inference starts them from pure noise. Including them with pad-token id as the CE target teaches the model to emit pad in the unused tail, which the decoder skips.

*BERT-initialized embeddings.* The token embedding table  $E$  is initialized from **bert-base-uncased** [3] rather than randomly, giving the rounding objective a meaningful geometry to start from. The embeddings are then trained jointly with the rest of the network.

*Loss.* We combine the embedding-space MSE with an auxiliary rounding cross-entropy on the target positions only:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{t, x_0, \epsilon} [\mathbf{1}_{\text{tgt}} \odot (\|x_0 - f_\theta(x_t, t)\|^2 + \text{CE}(W^\top f_\theta(x_t, t), w))] , \quad (2)$$

where  $W$  is tied to the input embedding matrix  $E$ .

#### 3.2 Self-Conditioning

At training time, with probability  $\frac{1}{2}$ , we first run a stop-gradient forward pass  $\tilde{x}_0 = f_\theta(x_t, t)$ , clamp its prefix positions back to  $x_0$ , and then take a second pass whose first hidden state is the sum of the usual input projection  $W_{\text{in}}x_t$ ,

a dedicated self-conditioning projection  $W_{sc}\tilde{x}_0$ , and the position and timestep embeddings. Gradients only flow through the second pass. On the other half of training steps,  $\tilde{x}_0$  is set to zero, which matches the test-time first denoising step. During sampling, the estimate  $\hat{x}_0$  from each DDIM step is detached and used as  $\tilde{x}_0$  at the next step.

### 3.3 Attribute Classifier and Guided Sampling

We jointly train a lightweight attribute classifier  $g_\phi : \mathbb{R}^d \rightarrow \mathbb{R}^K$  that operates on a target-masked pooled average of the noisy embeddings,

$$z = \frac{\sum_i m_i x_{t,i}}{\sum_i m_i}, \quad g_\phi(z) = W_2 \text{SiLU}(W_1 z), \quad (3)$$

where  $m$  is the binary target mask. The classifier is trained with cross-entropy against the discrete value of a chosen slot. For **food**, the label space is constructed once over the training split: index 0 is reserved for `<missing>` (the MR does not contain the slot), and the remaining  $K - 1$  indices enumerate the sorted set of observed **food** values. The classifier loss is added to the diffusion loss with weight  $\lambda=0.1$ :

$$\mathcal{L} = \mathcal{L}_{\text{diff}} + \lambda \cdot \text{CE}(g_\phi(z), y). \quad (4)$$

Crucially,  $g_\phi$  is trained on *noisy* pooled embeddings, exactly the input it will see at sampling time, so the classifier learns to score partially denoised states rather than only clean text.

At sampling, before each DDIM step we take a gradient ascent step on the target log-probability of the desired class  $y^\star$ :

$$x_t \leftarrow x_t + s \cdot \frac{\nabla_{x_t} \log p_\phi(y^\star | x_t)}{\|\nabla_{x_t} \log p_\phi(y^\star | x_t)\|_2}, \quad (5)$$

with guidance scale  $s=0.5$ . The gradient is zeroed out on the prefix positions, and the prefix is re-clamped after the update, so the MR remains untouched.

### 3.4 EMA Weights and Baselines

We maintain an exponential moving average of the denoiser parameters with decay 0.999 and use the EMA copy for validation and export, following standard diffusion practice [8,14].

**GPT-2** (small) [17]: causal LM fine-tuned on “MR <|sep|> reference” with the EOS token used as both pad and end. At decoding we use nucleus sampling with  $p = 0.95$  and temperature 1.0, which trades a small BLEU drop for higher distinct- $n$  — the diversity-oriented setting that gives diffusion the strongest possible baseline to match on lexical variety. **T5-small** [18]: seq2seq with source=MR, target=reference; beam search with  $b=4$ . T5 with beam search is the strongest pure fluency baseline.

## 4 Experimental Setup

*Dataset.* E2E NLG cleaned [15,5]; 33,525 / 4,299 / 4,693 train/val/test pairs. MR slots: **name**, **eatType**, **food**, **priceRange**, **customerRating**, **area**, **familyFriendly**, **near**. MRs are flattened to a “key: value | key: value” string before tokenization with the **bert-base-uncased** WordPiece tokenizer.

*Training.* All models are trained on a single A100 GPU (Google Colab Pro+). **Diffusion-LM**: 20 epochs, batch size 64, AdamW, learning rate  $1 \times 10^{-4}$ , linear warmup over 1000 steps, gradient clipping at norm 1.0,  $T=2000$  diffusion steps. **Baselines**: 5 epochs, batch size 32; learning rates  $5 \times 10^{-5}$  (GPT-2) and  $3 \times 10^{-4}$  (T5). Maximum sequence length is 96 tokens with prefix length 32.

*Decoding.* GPT-2: nucleus sampling ( $p=0.95$ ,  $\tau=1.0$ ). T5: beam search with  $b=4$  and length penalty 1.0. Diffusion-LM: DDIM [19] with 200 steps and a rounding interval of 10 steps, self-conditioning enabled, classifier guidance scale 0.5 on the **food** slot. Generation is batched (16 examples per batch) to keep inference cost tractable.

*Metrics.* *Fluency*: BLEU [16] via **sacrebleu**, ROUGE-L [12] (F1). *Diversity*: distinct-1 and distinct-2 [9], the ratio of unique  $n$ -grams to total  $n$ -grams in the generated corpus. *Controllability*: slot-fill accuracy, the fraction of MR slot values that appear verbatim (case-insensitive) in the generated output. Slot-fill accuracy is a permissive lexical-match metric — it does not penalize paraphrase but does catch slot omissions, which is the failure mode E2E benchmarks have been historically designed to surface.

*Reproducibility.* All configurations live in YAML files under **configs/**; a 10-second CPU smoke test covers parsing, all three schedules, forward/backward/sample with prefix clamping, self-conditioning, classifier guidance, and the full metric pipeline. Training writes both raw and EMA checkpoints per epoch and selects best-val-loss for evaluation.

## 5 Results

Table 1 reports the test-set comparison across the three systems; Table 2 reports the Diffusion-LM ablations on noise schedule, DDIM step count, and prefix length. Sections 6 and 7 interpret these in turn.

## 6 Analysis

*Expected slot-level failure modes.* The **food** and **near** slots have the largest open vocabularies in E2E and are known to be the hardest to recover lexically [5]: they are most often paraphrased (“Indian”  $\rightarrow$  “curry house”) or omitted entirely. We therefore expect T5 with beam search to be the most slot-faithful overall but

**Table 1.** Main results on E2E NLG test set (4,693 examples). BLEU and ROUGE-L are reported on a 0–100 scale; distinct- $n$  on a 0–1 scale.

| Model               | BLEU | ROUGE-L | distinct-1 | distinct-2 | Slot Acc. |
|---------------------|------|---------|------------|------------|-----------|
| GPT-2 (nucleus)     | ...  | ...     | ...        | ...        | ...       |
| T5-small (beam-4)   | ...  | ...     | ...        | ...        | ...       |
| <b>Diffusion-LM</b> | ...  | ...     | ...        | ...        | ...       |

**Table 2.** Ablations on Diffusion-LM. The sqrt schedule with prefix length 32 and 200 DDIM steps is the default configuration reported in Table 1. Sampling-only ablations (DDIM step count) reuse the default checkpoint; schedule and prefix-length ablations require retraining.

| Variant                 | BLEU | ROUGE-L | distinct-2 | Slot Acc. |
|-------------------------|------|---------|------------|-----------|
| sqrt schedule (default) | ...  | ...     | ...        | ...       |
| cosine schedule         | ...  | ...     | ...        | ...       |
| DDIM 50 steps           | ...  | ...     | ...        | ...       |
| DDIM 100 steps          | ...  | ...     | ...        | ...       |
| DDIM 200 steps          | ...  | ...     | ...        | ...       |
| prefix length 16        | ...  | ...     | ...        | ...       |
| prefix length 32        | ...  | ...     | ...        | ...       |
| prefix length 48        | ...  | ...     | ...        | ...       |

the least lexically varied, GPT-2 with nucleus sampling to be the most varied but prone to hedging on contradictory MRs (e.g. `priceRange[high]` with `customerRating[low]`), and Diffusion-LM with classifier guidance to raise the `food-slot` accuracy specifically while leaving the other slots controlled by prefix clamping alone. These predictions are verified against the per-example outputs and broken down by slot in the companion notebook `notebooks/error_analysis.ipynb`.

*Classifier guidance scale.* The guidance scale  $s$  trades off attribute fidelity against on-manifold fluency: at large  $s$  the gradient update pulls the denoiser away from the training distribution, hurting BLEU; at very small  $s$  the update is indistinguishable from no guidance. We selected  $s = 0.5$  via a coarse validation grid search before reporting test-set numbers; we do not tune  $s$  further on the test split.

## 7 Discussion

*Hard vs. soft conditioning and the diversity–fluency frontier.* Prefix clamping is hard conditioning — the MR is never noised and never moved — and is the right choice for slots that must appear verbatim (e.g. `restaurant name`). Classifier guidance is soft and is the right choice for slots that benefit from paraphrase (e.g. `food`: “Italian”, “Italian cuisine”, “Italian dishes”). Our implementation supports both simultaneously. Along the diversity–fluency axis the three systems

are expected to occupy distinct points: T5 beam search optimizes likelihood and collapses to templates, GPT-2 nucleus sampling trades a small BLEU drop for diversity, and Diffusion-LM sits in between, with the rounding step pulling toward high-probability tokens at every iteration.

*Limitations.* Three constraints shape our results. Our denoiser has six layers and  $\sim 50\text{M}$  parameters, an order of magnitude smaller than the original Diffusion-LM; at this scale, performance is bounded by capacity rather than algorithm. E2E NLG is heavily templated — short references ( $\sim 20\text{--}30$  tokens), mostly-closed slot vocabulary, naturally low reference diversity — which structurally favors beam search; a benchmark with multiple valid outputs and more open vocabulary (e.g. CommonGen [11], ROCStories [13]) would make the diffusion advantages more visible. Finally, slot-fill accuracy is a permissive lexical-match metric and does not credit paraphrase; a parser-based MR-recovery score would be more faithful.

*Ethical considerations.* Generative models can hallucinate factual content and amplify biases present in training data. On E2E the risk is bounded (synthetic restaurant descriptions in a controlled vocabulary), but the framework is general: deployed at scale on open-domain text it would require bias auditing and content filtering. We do not release a trained checkpoint with this work.

## 8 Conclusion

We presented prefix-conditioned Diffusion-LM for controllable text generation on E2E NLG, augmented with self-conditioning, EMA, BERT-init, and a jointly trained attribute classifier for gradient-based guidance during DDIM sampling. The model combines hard prefix conditioning, which enforces attribute control by construction, with soft classifier guidance, which trades fluency for finer per-slot control, and was compared with GPT-2 and T5 baselines along fluency, diversity, and controllability. Three next steps follow from our limitations: a multi-head classifier guiding all eight slots simultaneously, a parser-based MR-recovery metric that credits paraphrase, and evaluation on a benchmark whose structure rewards iterative refinement — e.g. CommonGen, ROCStories — where the diffusion paradigm has more room to differentiate from beam search.

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